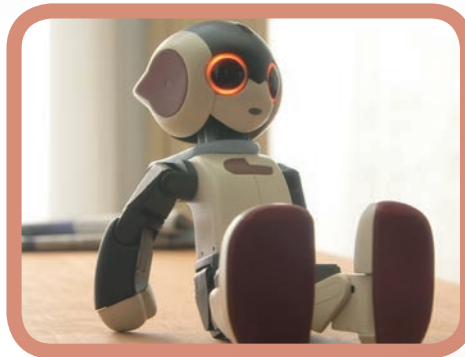
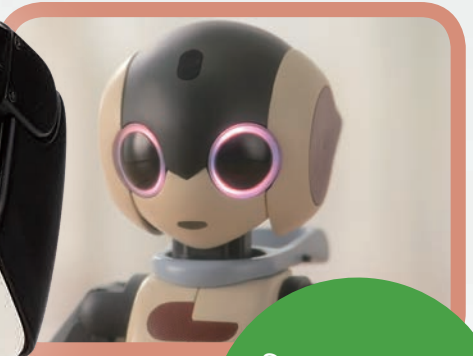


Build your own Robi

Pack 07



Stages 23-26:
Begin assembling
Robi's right foot



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Assembly Guide

p120

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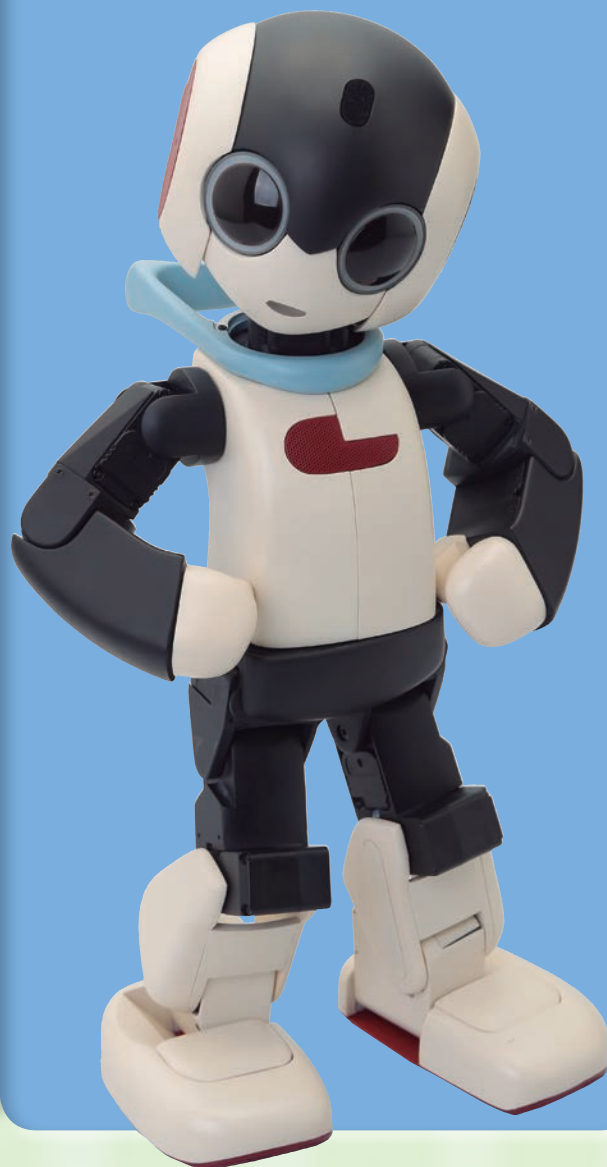
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COLOUR VARIATIONS OWING TO THE MATERIALS USED.



STAGE 23: PREPARE ROBI'S FIRST FOOT SERVO

In this assembly, you will prepare, test and set the servo motor that will operate the side-to-side tilt of Robi's right foot, servo ID=6.

As you did with the servos that make up Robi's arms, you will begin the work on his feet and legs with the servo farthest from his body, in this case the servo that will tilt Robi's right foot from side to side. The work is the same as for his other servos, but it is vital that great care is taken as this is done, as it involves exposing the delicate electronic parts within.



YOUR PARTS

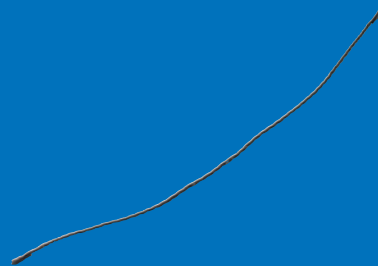
1 Servo motor

**PARTS TO BE
ASSEMBLED**



SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.



200mm servo cable (Stage 22)



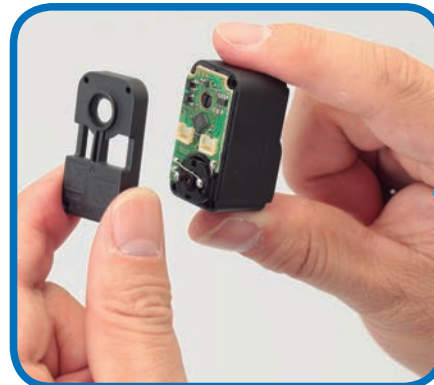
Neck stand (Stage 8)

THE SERVO AND CABLE

TIP!

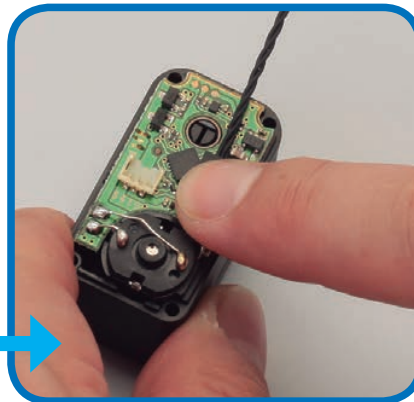


As ever, it is very important that the electronic parts exposed by removing the servo's cover are not touched in any way. Make sure to hold the open servo by its sides, and if you put it down on your work surface during the assembly, do this with the open side facing upwards.



1 Take out the four screws holding the servo's removable cover in place, and keep these safely to one side.

2 Gently lift the cover away, taking care not to force it or damage the circuit board underneath.



TIP!



This servo cable had protective pads fitted to its connectors in the previous stage, but if you find it difficult to attach with the pad in place, it is possible to apply the pad afterwards. If you choose to do the assembly this way, be especially careful not to press down on the circled projection, next to the cable, as you do so.

3 Hold the open servo with the circuit board facing upwards. Fetch the servo cable fitted with protective pads in the previous stage, and press one of its connectors into the socket circled in red, as you have done before. It does not matter which socket, or which end of the servo cable you use.



4 Pass the free end of the servo cable through the circular hole in the removable cover, as shown.

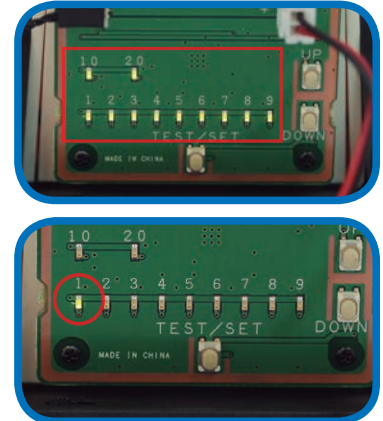
5 Press the cover back into place on the servo's main housing.

6 Fasten the cover back in place with all four screws, but only lightly, as you will be removing them again at a later stage.

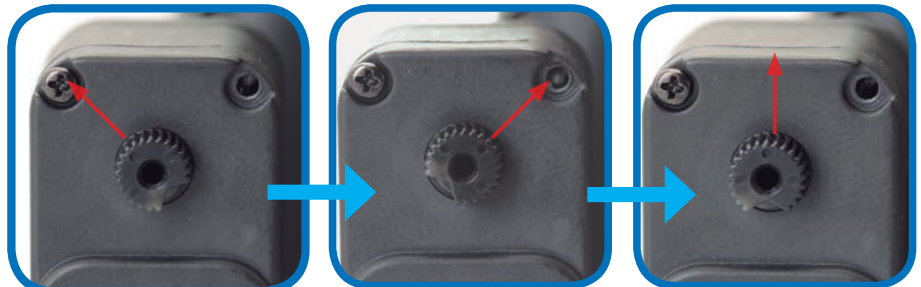
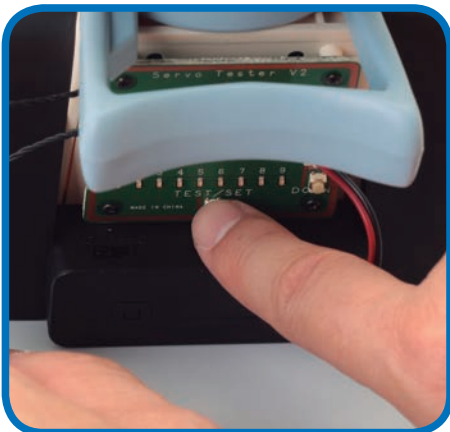
TESTING THE SERVO



- 7** Take the neck stand and make sure the power pack is turned OFF. Carefully unplug the black servo cable from the socket on the left of the test board and attach the servo cable connected to the servo in Step 3.



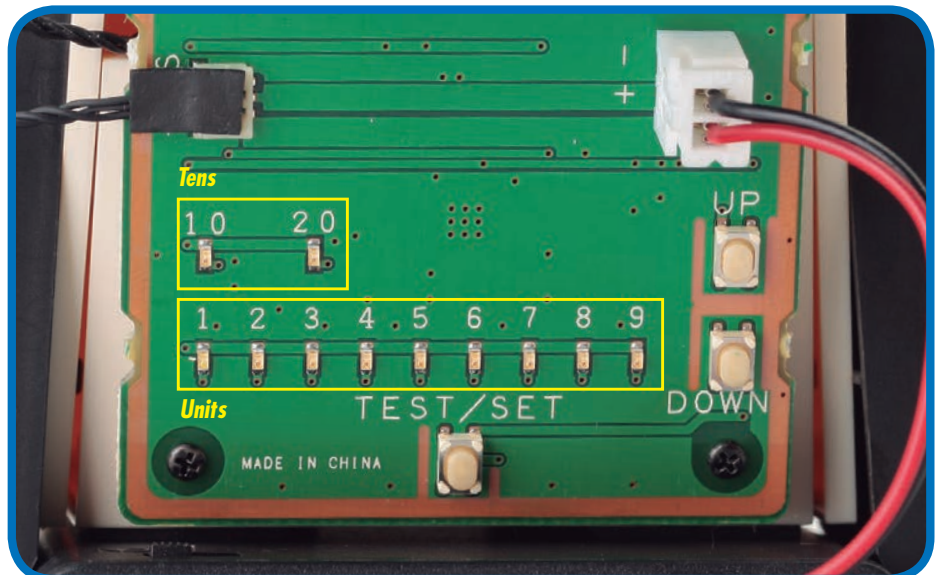
- 8** With the servo connected, switch the power to ON. After the LEDs blink twice, only the 1 LED should light up.

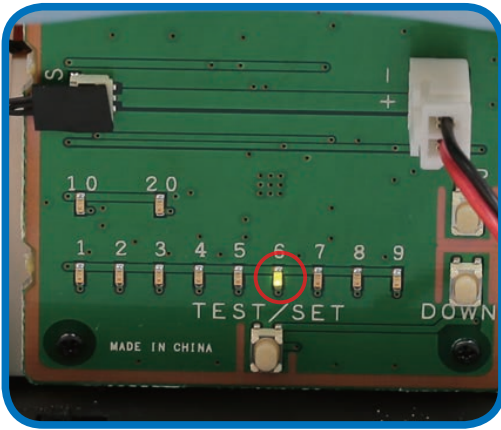


- 9** To test the servo, press the TEST/SET switch: the servo's shaft should rotate 45 degrees to the left, stop, then move back across to 45 degrees to the right. The next step is to set the servo's ID number, so keep the power on. If the test does not run, or if the LEDs flash rapidly, refer to the HELP box in Stage 8.

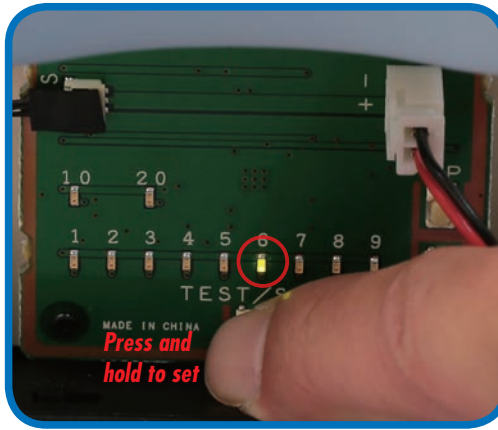
SETTING THE ID NUMBER

- 10** Re-familiarise yourself with the lights and switches on the test board. These are used to set the ID number of the servos. Each time you press the UP switch, the ID number increases by one, and the result is displayed by the two banks of LED lights. The top row reads in tens and the bottom in units. For example, to set the ID to 19, you would press UP 18 times. This will light up the 10 LED and the 9 LED. (If you overshoot, simply press the DOWN switch to take it down one step at a time.) When the ID number is correct, you set it by pressing and holding the TEST/SET switch.





- 11** To set the ID number of the right foot servo to 6, press the UP button five times. Make sure that only the 6 LED is lit, using the UP and DOWN buttons to correct the number if necessary.



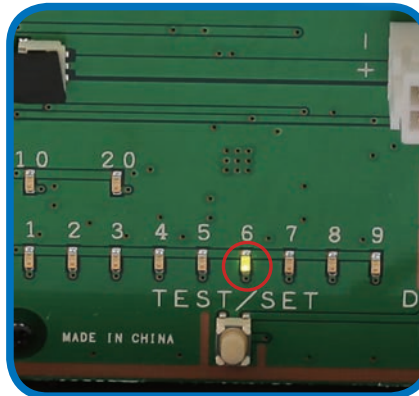
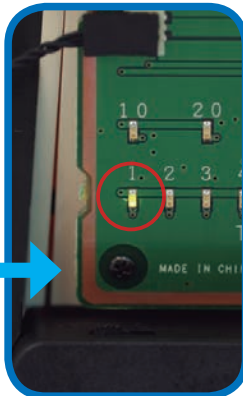
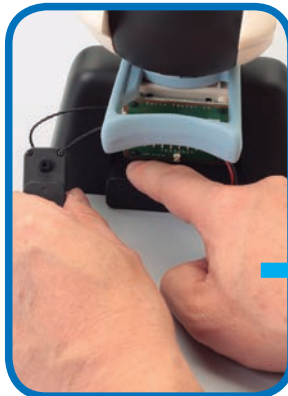
- 12** Confirm the setting by pressing and holding the TEST/SET button for a few seconds. The 6 light should flash a few times, then remain lit. The ID number is now set.



ID=6

CHECKING THE ID NUMBER

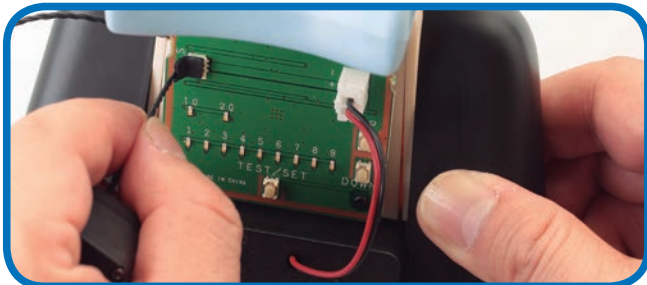
HELP!



It is normal for the 1 LED to light up when you switch on. It will only change to give you the ID number of the servo when you press TEST/SET. If you made a mistake or set the wrong ID number, you can reset it by going through the setting procedure again.

- 13** It is vital that Robi's servos are set correctly, so it is always a good idea to check that you have set the intended ID number afterwards. To do this, leave the servo connected, switch the power OFF, then back ON. After all the LEDs blink, only the 1 LED should light up.

- 14** To check the ID number, press TEST/SET. Now only the 6 LED should light up. Note: if you find that a servo has the wrong number assigned to it, go back and repeat the setting process in full.



- 15** Once all is well, switch the power pack OFF, then very carefully unplug the servo from the test board. Pull the servo cable away gently, directly upwards from the socket (do not pull from an angle, as this can damage the connector). This is especially important if you chose not to add the protective pad earlier.

Assembled foot servo

The servo that will tilt Robi's right foot is assembled, wired and set. Make sure you keep this part safely, preferably in a sealed plastic bag with its ID number (6) written on it for ease of reference.



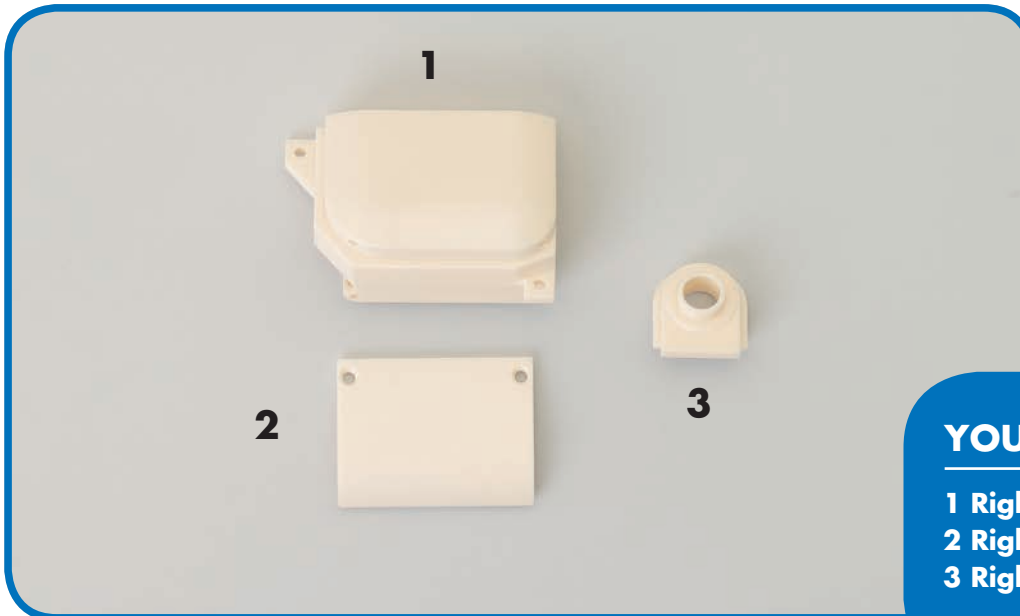
STAGE 24: FITTING THE FOOT SERVO COVER

In this assembly you will fit the servo prepared in Stage 23 (ID=6) to its cover, as Robi's right foot starts to take shape.

The servo supplied and prepared in the previous stage, ID=6, will tilt Robi's right foot from side to side on completion. In this stage you will fit it to its cover, which is coloured and moulded to fit the rest of Robi's

exoskeleton. Along with the cover that you will use in this build, you have received other parts that will make up Robi's right ankle – you will not be using these now, so be sure to keep them safely until they are needed.

**PARTS TO BE
ASSEMBLED**

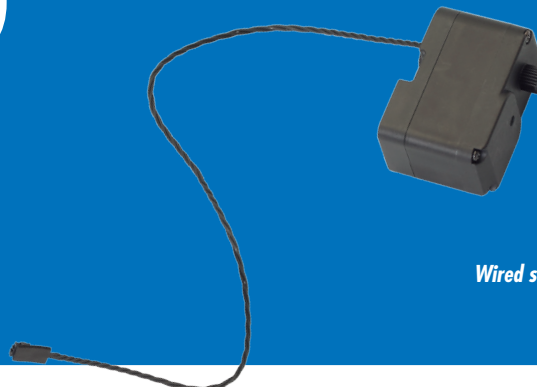


YOUR PARTS

- 1 Right foot cover
- 2 Right ankle side panel
- 3 Right ankle joint

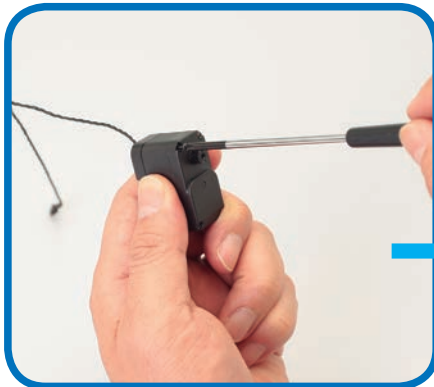
SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.

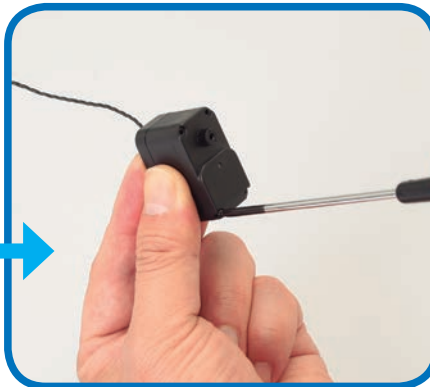


Wired servo (Stage 23)

PREPARE THE SERVO

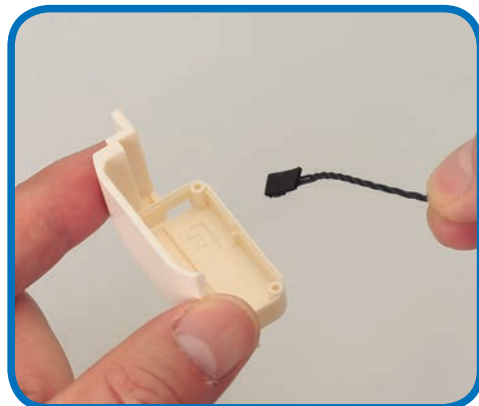


- 1** Take out the four screws holding the servo's removable cover in place, and keep these safely to one side.

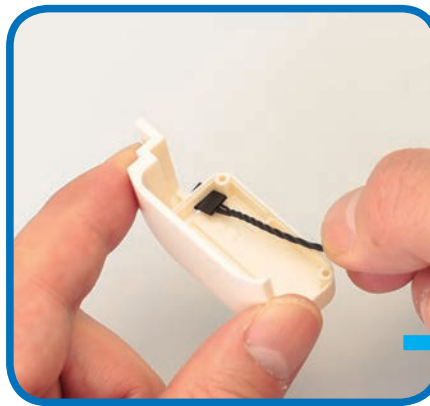


- 2** Gently lift the cover away, taking care not to force it or touch the circuit board underneath.

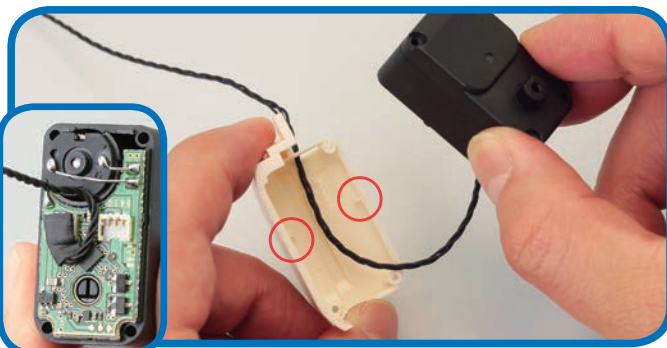
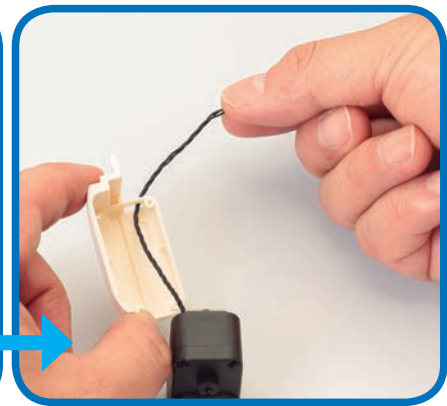
THE RIGHT ANKLE COVER



- 3** Line up the free end of the prepared servo's cable to the right ankle cover, as shown.



- 4** Carefully feed the servo's connector through the rectangular hole in the cover, as shown. Do not pull too hard as you do this, and do not let the cable get caught. Stop when about two thirds of the cable is through.



- 5** Lower the open-faced servo into the cover, feeding the cable through the hole as you do so. Be very careful that the cable does not bunch up inside the cover, and that it does not get caught on the two circled projections, as this may damage it. Turning the cable neatly over on itself before lowering the servo into place (inset) will help avoid this.



- 6** With the cable neatly folded and leading through the hole as shown, lower the servo fully to the cover. Keep the cable slightly taut as you do this, to avoid any slack forming inside the cover.



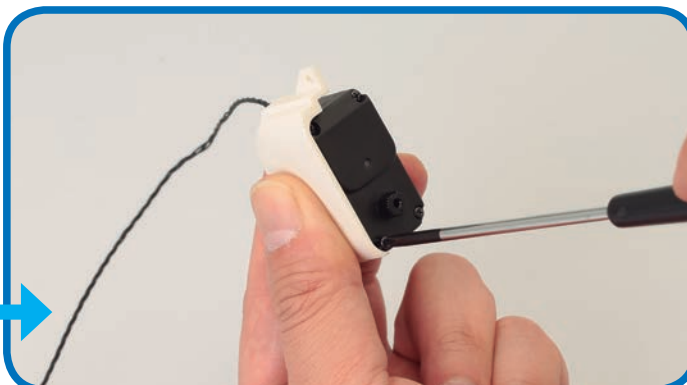
7 Press the servo carefully into place.



8 Check that the servo and cover are meeting neatly along their edges, and that no gaps or evidence of the cable having been trapped or pinched can be seen.



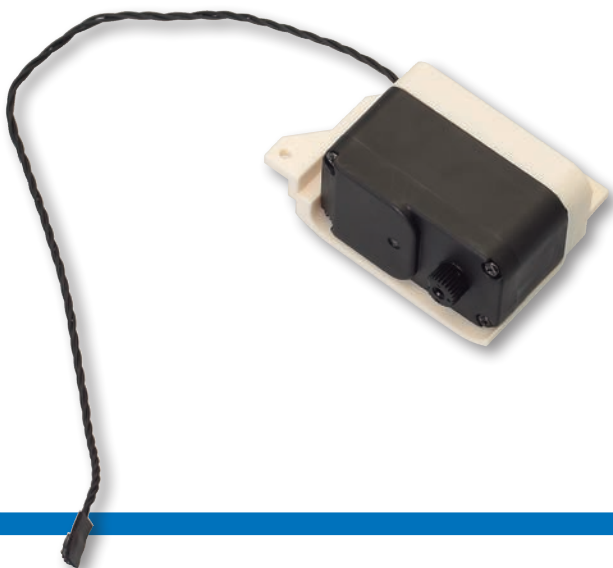
9 Use the four screws removed in Step 1 to fasten the servo to its cover.



TIP!

Assembled foot servo

The servo that will tilt Robi's right foot from side to side is wired and fitted to its cover. Make sure you keep this subassembly safely, ideally in a sealed plastic bag with its ID number (6) written on it for ease of reference.



STORAGE OF PARTS



In this stage, you have received parts that are not used in this assembly's steps. These should be stored neatly away, ideally in a compartmentalised hobby box, as mentioned in the previous pack. It is also advisable that you keep all Robi's as-yet unused plastic parts out of direct sunlight, as well as separated by colour, to preserve their appearance until they are needed.

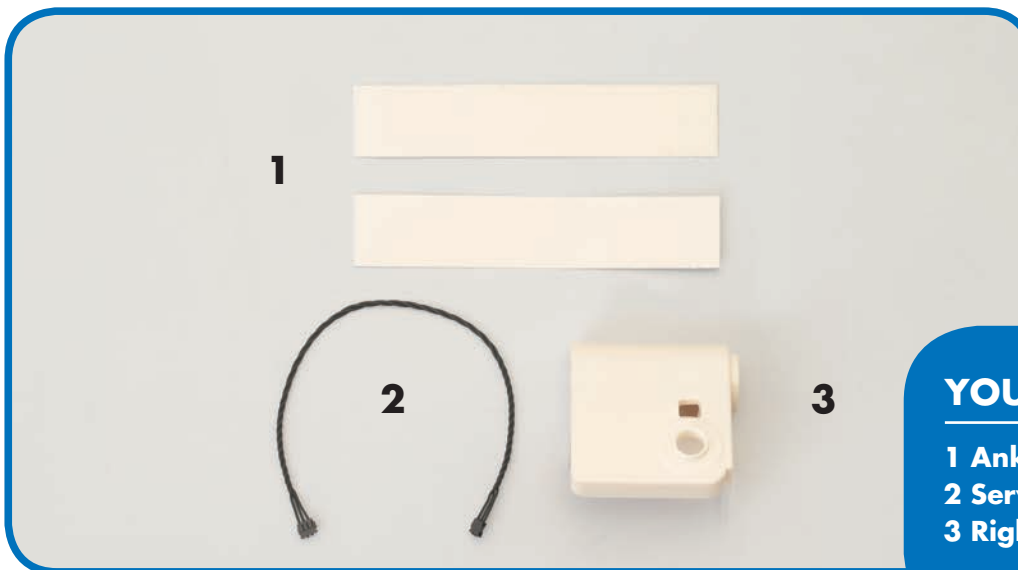
STAGE 25: JOINING THE FOOT SERVO TO THE ANKLE FRAME

In this assembly, you will join the foot cover and servo subassembly built in the previous stage to the ankle frame, ready to connect the ankle servo in the coming stages.

Robi's foot servo will work in tandem with his ankle servo, the side-to-side tilting motion of the former working with the up-and-down rotation of the latter, giving him the flexibility needed to walk naturally. In this stage you prepare to connect these two servos,

by joining the frame that will hold Robi's ankle servo to the shaft of the right foot servo (ID=6) from the previous stage. The ankle tape received with this stage will not be used until next time, so keep this carefully until then.

PARTS TO BE ASSEMBLED IN RED



YOUR PARTS

- 1 Ankle tape
- 2 Servo cable (135mm)
- 3 Right ankle frame

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.

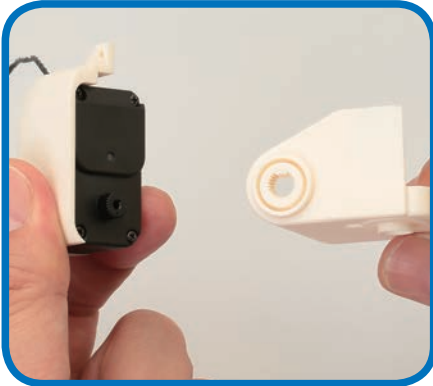


Servo and foot cover subassembly (Stage 24)

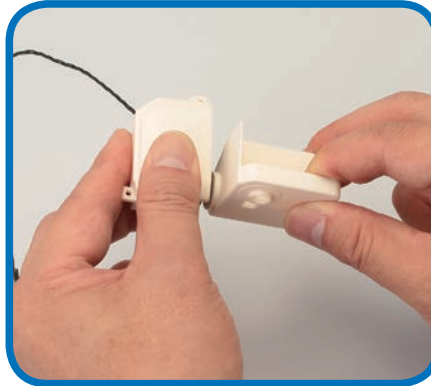


Protective pads (Stage 3)

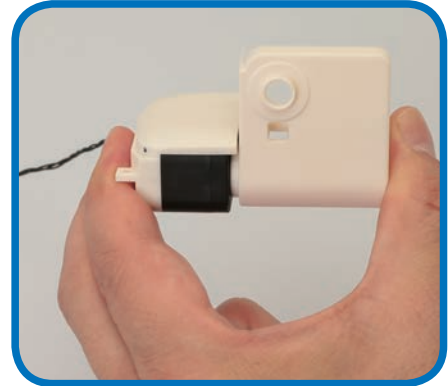
JOINING THE FOOT SERVO TO THE ANKLE FRAME



- 1 With the foot servo and ankle frame positioned at right angles as shown, line up the servo's shaft to the hole in the ankle frame. Note the location of the notch in the servo's shaft, which should line up exactly to the notch in the frame's D-cut hole.



- 2 With the parts lined up exactly, press the ankle frame onto the servo's shaft, making absolutely sure not to force the parts (see the Close Up box below).



- 3 Check that the parts are joined neatly, and that no gap can be seen at the point where the shaft enters the frame. The parts should be positioned at right angles to one another.

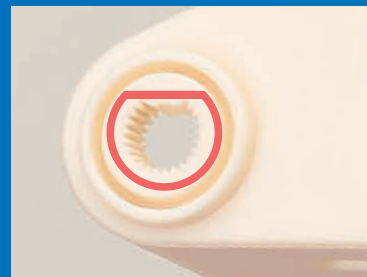
HANDLE WITH CARE!



This stage sees you working with D-cuts once again. As with the previous instances of these, care must be taken to ensure that the correspondingly shaped parts are fitted together properly, as this will ensure the correct 'neutral' position is achieved, and avoid irreversible damage.

Locate the D-cuts

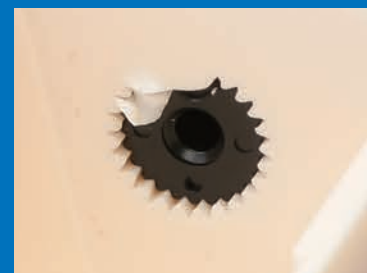
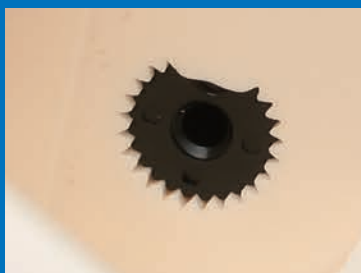
The servo shaft is not fully cylindrical, but has a small, triangular notch in it. This creates the outline of a letter 'D', as shown in the photo near left. To match this, the corresponding hole in the ankle frame has a small flat section in it, again creating the outline of a letter 'D' (far right).



Along with the smaller teeth used to grip each other, the parts have corresponding notches and indents designed to ensure the correct neutral position is achieved.

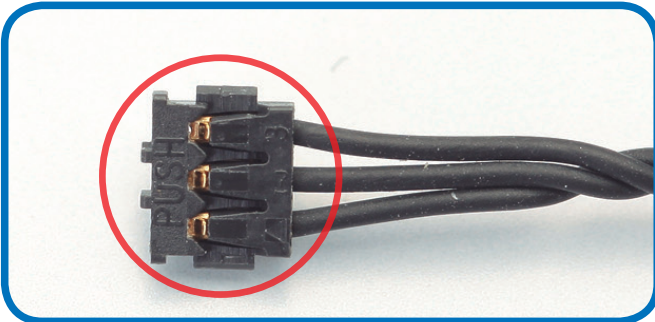
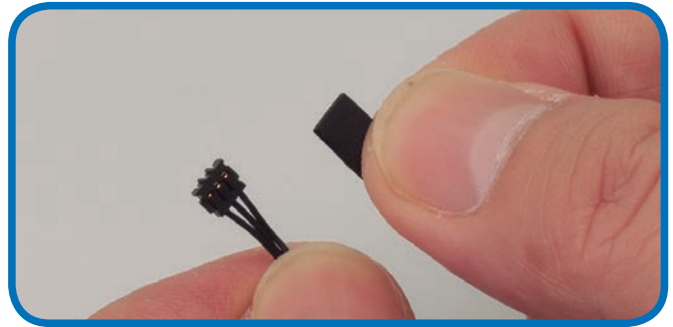
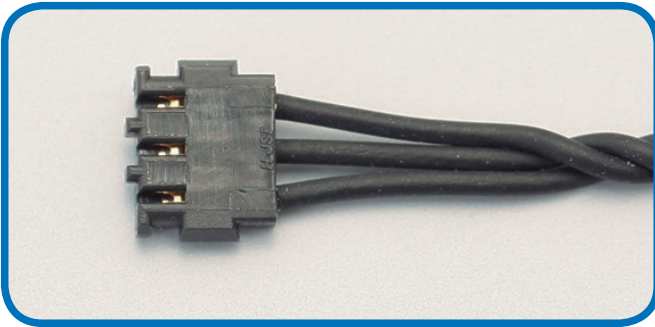
A clean fit

The frame's hole is designed to match the servo shaft, with the D-cut outline of the shaft fitting to the frame exactly. It is vital that the parts connect neatly, as severe damage can occur otherwise.



The frame should fit over the servo's shaft neatly and easily. Never force it, as this may well result in irreparable damage to the frame, as seen in the photo near right.

SEALING THE SERVO CABLE



- 4** As you have done previously, identify the side of this stage's servo cable's connector that has PUSH marked on it (circled).

- 5** Apply a protective pad to the connector on this side. Repeat for both ends of the cable.

TIP!

During the build so far, you may have found it easier to apply the protective pads after fitting the cable to the servos' sockets. If this is the case, you can leave the cable's connectors uncovered for now, and fit the pads later.



Assembled foot servo and ankle frame

The servo that will tilt Robi's right foot from side to side is now joined to the frame that will hold Robi's ankle servo. Make sure you keep this subassembly safely until it is next needed.

STORAGE OF PARTS



The ankle tape can be stored along with the side panel and joint supplied in the last stage, as they will be used in the coming assemblies.

STAGE 26: PREPARE ROBI'S ANKLE SERVO

In this assembly you will prepare, test and set the ID number of the ankle servo that will operate the up-and-down action of Robi's right foot, servo ID=5. This will include wrapping the servo's body in coloured ankle tape.

The next servo that will operate Robi's right foot is servo ID=5, his ankle servo. This fits into the ankle frame fitted in the last stage, to combine with Robi's right foot servo to bring his foot up and down, enabling

him to walk. Once you have wired and set the ID number of the servo, you will wrap it with the white ankle tape supplied in Stage 25, which will ensure the part fits Robi's colour scheme on completion, so have this to hand.



YOUR PARTS

1 Servo motor

PARTS TO BE ASSEMBLED



SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.

135mm servo cable
(Stage 25)



Ankle tape x 1
(Stage 25)



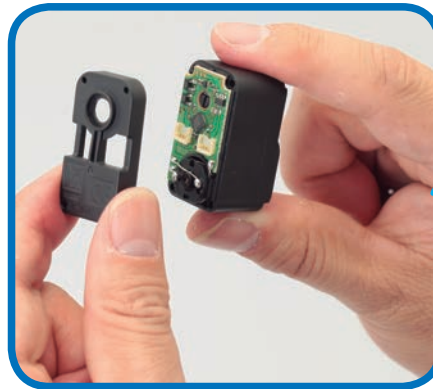
Neck stand (Stage 8)

THE SERVO AND CABLE

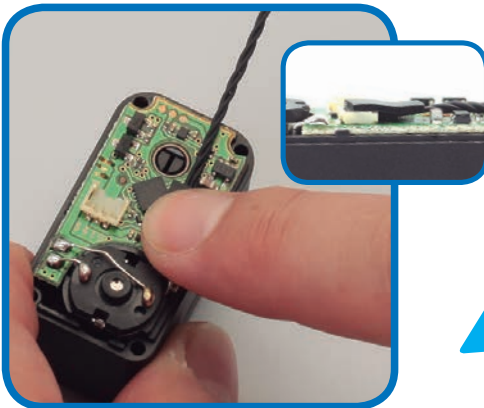
TIP!



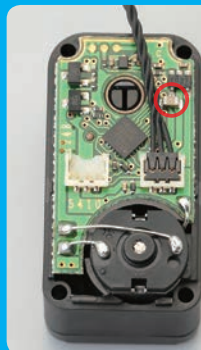
As ever, it is very important that the electronic parts exposed by removing the servo's cover are not touched in any way. Make sure to hold the open servo by its sides, and if you put it down on your work surface during the assembly, do this with the open side facing upwards.



- 1 Take out the four screws holding the servo's removable cover in place, and keep these safely to one side. Then gently lift the cover away, taking care not to force it or damage the circuit board underneath.



TIP!

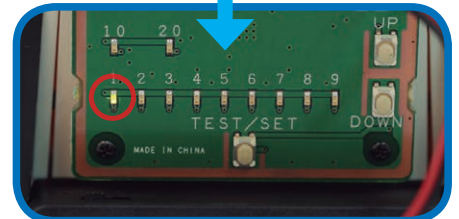
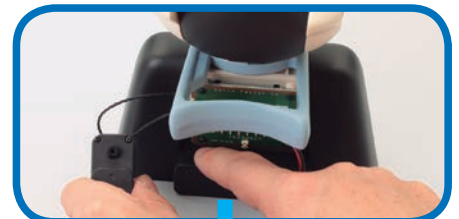
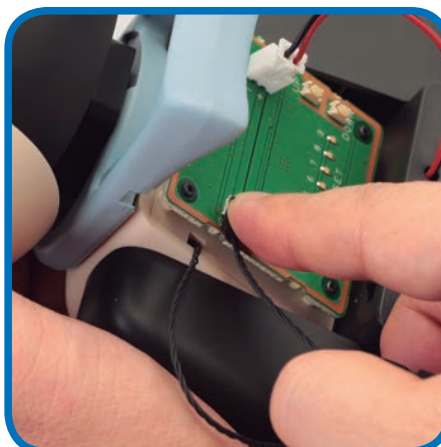


If you find it difficult to attach the cable with the pad in place, it is possible to apply the pad afterwards. If you choose to do this, be especially careful not to press down on the circled projection, next to the cable, as you do so.



- 2 Fetch the servo cable fitted with protective pads in the previous stage, and press its connector into one of the servo's sockets, as you have done before. Make absolutely sure that the connector is set into place correctly, and that its protective pad is sitting flush to the circuit board, as shown.

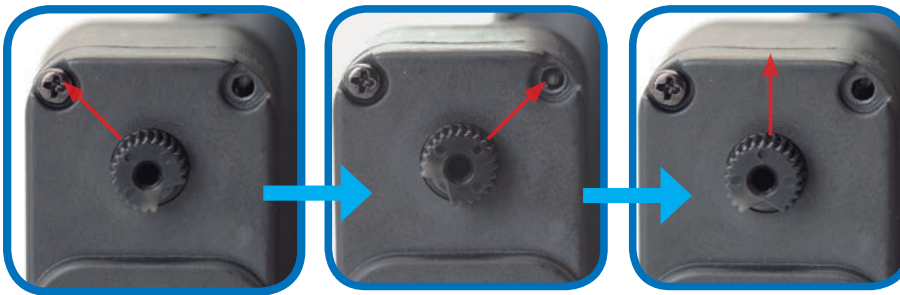
- 3 Pass the free end of the servo cable through the circular hole in the removable cover, as shown.



- 4 Replace and secure the cover with all four screws, but only lightly as you will be removing them again at a later stage.

- 5 Making sure the power is turned OFF on the test board, unplug the existing cable from its socket.

- 6 Attach the free connector of this stage's servo, and turn the power ON. The LEDs should flash twice, then 1 will remain lit.



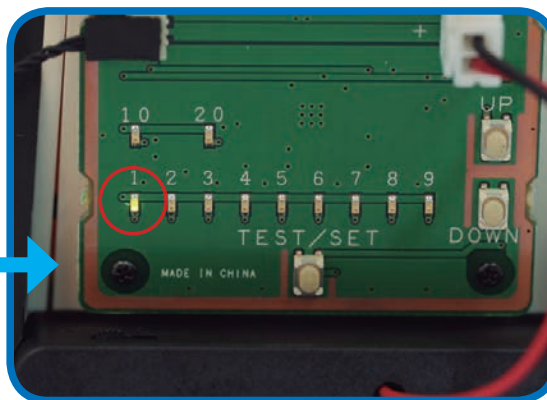
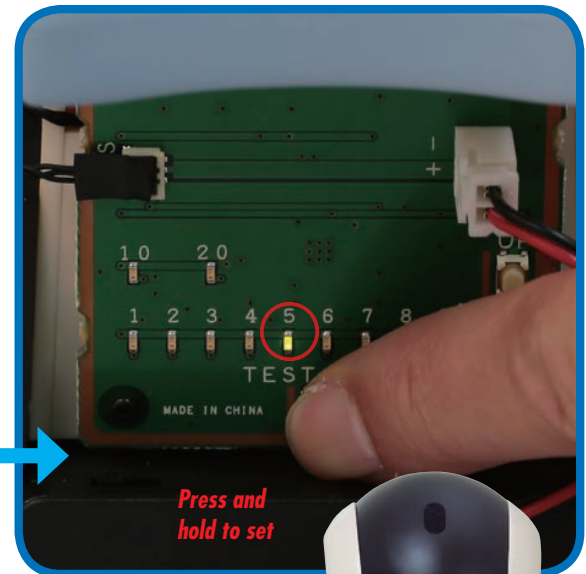
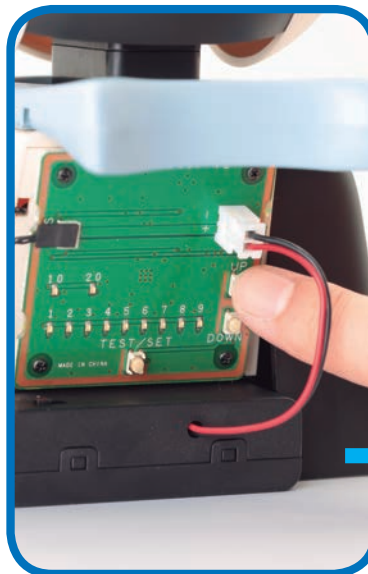
TIP!

If the test does not run when you press the TEST/SET button, or if the LEDs flash rapidly, you may have a faulty connection. Check that the connector is fitted correctly into the test board's socket, and if it is, carefully unscrew the cover of the servo (replaced in Step 4) and check the connection there. For more info on this, refer to the HELP box in Stage 8.

- 7** To test the servo, press the TEST/SET switch: the servo's shaft should rotate 45 degrees to the left, stop, then move back across to 45 degrees to the right. The next step is to set the servo's ID number, so keep the power on.

SETTING THE ID NUMBER

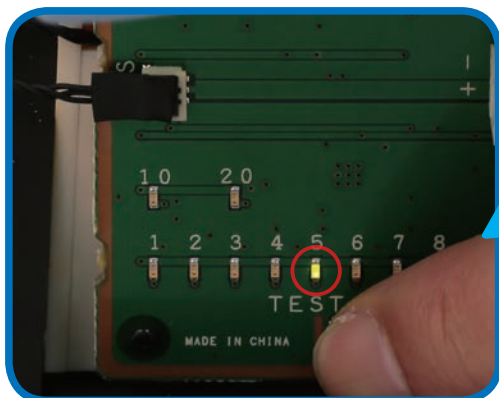
- 8** By now you will be familiar with the process of setting servos' ID numbers, but refer back to previous stages' steps if you need a reminder. The servo being prepared in this stage will be ID=5, so press the UP button four times, so that only the 5 LED is lit. When this is the case, press and hold the TEST/SET button for a few seconds. The 5 LED should blink, then remain lit.



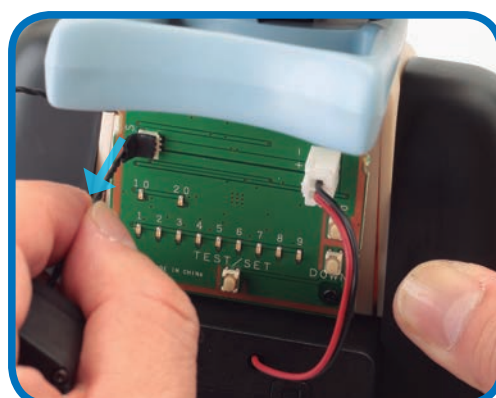
- 9** It is vital that Robi's servos are set correctly, so it is always a good idea to check that you have set the intended ID number afterwards. To do this, leave the servo connected, switch the power OFF, then back ON. After all the LEDs blink, only the 1 LED should light up.

ID=5





It is normal for the 1 LED to light up when you switch on. It will only change to give you the ID number of the servo when you press TEST/SET. If you made a mistake or set the wrong ID number, it is essential that you reset it by going through the setting procedure again.



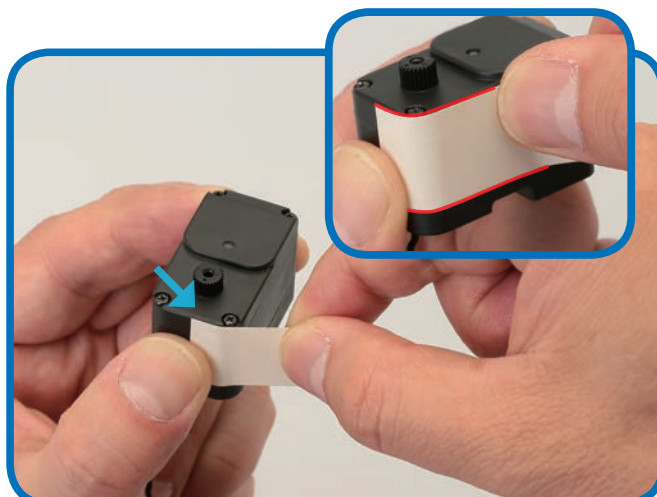
- 10** To check the ID number, press TEST/SET. Now only the 5 LED should light up. Note: if you find that a servo has the wrong number assigned to it, go back and repeat the setting process in full.

- 11** Once you are absolutely certain that the servo is set correctly, turn the power OFF and carefully remove the servo cable from the test board. Pull the cable away gently, directly upwards from the socket (never pull at an angle). This is especially important if you have not yet applied a protective pad to the connector.

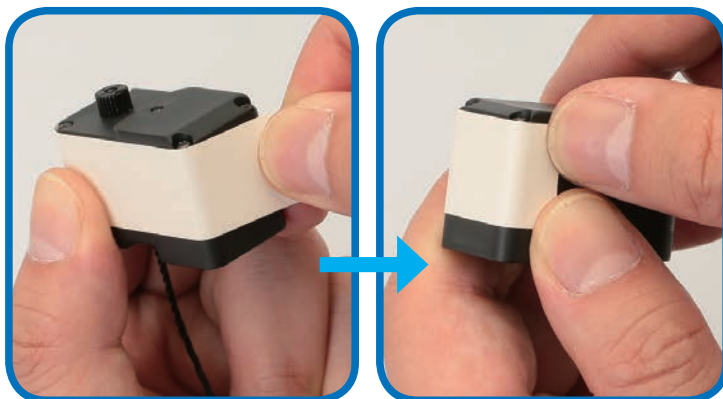
APPLYING THE ANKLE TAPE



- 12** Take one strip of ankle tape saved from Stage 25 (you only need one of them in this stage, but keep the other safely). Carefully peel the tape away from its backing.



- 13** Apply the tape, sticky side down, to the side wall of this stage's servo, starting near the shaft. Now start wrapping the tape around the servo's body, following the photo exactly. Make sure no part of the tape extends beyond the edge of the servo (marked by the red lines, inset). It is very important that the tape does not overlap onto the servo's face in any way, as this will inhibit the operation of the moving parts that will attach there.



- 14** Still making sure no tape extends beyond the side of the servo, keep wrapping the tape around until it is finished.

Prepared ankle servo

Robi's right ankle servo is now set with its ID number (ID=5), is wired, and has its tape applied, ready to be fitted in coming stages.



COMING IN PACK 8

Your next **FOUR** complete Stages, as Robi starts to come to life!



Stage 27

Add the right foot cover and sole

PARTS PROVIDED

More parts of Robi's right foot, including the sole.



Stage 28

Add the right lower leg frame and cover

PARTS PROVIDED

The outer lower leg frame and cover that will join the ankle and knee servos.



Stage 29

Prepare Robi's right knee servo

PARTS PROVIDED

The servo that will operate Robi's knee.



Stage 30

Add Robi's right knee frame

PARTS PROVIDED

The next parts of Robi's right leg, this time the inner lower leg and knee frames.

